

# Sachin Kumar

macOS Systems & Endpoint Security Engineer | iOS Developer

Bengaluru, Karnataka, India | +91 9501841073 | sachinmehtab@gmail.com

linkedin.com/in/sachinkumar6174 | github.com/sachin6174 | sachinserver.in | leetcode.com/u/sachinkumar6174

## SUMMARY

macOS systems engineer with 3+ years building the **SureMDM** endpoint-management agent at 42Gears, deployed across **10,000+ managed enterprise devices**. Specialized in privileged system agents (**LaunchDaemon**, **XPC**, helper tools), **NetworkExtension** traffic filtering and proxying, data-at-rest encryption with **CryptoKit** and **Keychain**, **Just-In-Time privilege escalation**, and location- and time-based compliance policy enforcement. Owns the trusted-distribution chain end to end: **code signing**, **notarization**, and **PKG/DMG** packaging.

## WORK EXPERIENCE

### 42Gears Mobility Systems Pvt. Ltd.

February 2023 – Present

Bengaluru, India | Unified Endpoint Management (UEM) SaaS

#### iOS & macOS Developer — SureMDM Agent for macOS

July 2023 – Present

- Architected privileged macOS system agents using **LaunchDaemon** process daemonization, **XPC** inter-process communication, and privileged helper tools, running as root across **10,000+ enterprise endpoints**.
- Built **NetworkExtension** providers (**NEFilterProvider** content filtering, **NEAppProxyProvider** flow proxying) delivering real-time network **traffic filtering** and per-application data usage attribution on managed devices. *[CONFIRM: mention TCP/UDP flow handling if you worked at flow level]*
- Implemented **Just-In-Time (JIT) privilege access** for on-demand, time-bounded privilege escalation, replacing standing administrator rights and aligning with least-privilege enterprise security policy.
- Led a data-at-rest security migration, moving unencrypted sensitive local storage to encrypted formats via **CryptoKit** and **Keychain Services**, and retiring deprecated third-party libraries in favour of first-party Apple SDKs with full backward compatibility.
- Engineered location-aware and schedule-based compliance enforcement — **Core Location** with reverse geocoding for geofenced policy, plus **Time Fence** restricted execution windows — to satisfy enterprise data-compliance requirements in restricted zones.
- Built enterprise **app lifecycle management** (PKG/DMG install, upgrade, and uninstall) combining Swift, **Bash** scripting, and **XPC**, replacing manual IT provisioning across the fleet.
- Owned the trusted distribution chain: **code signing**, **notarization**, **PKG/DMG** packaging, entitlement configuration, and automated build distribution through **GitLab CI/CD**.
- Developed device monitoring and remote administration modules covering hardware state changes, uptime tracking, user management, and administrator-triggered reboot and shutdown actions.
- Engineered a resilient **Download Manager** with resumable chunked transfers and bounded concurrent worker queues for secure payload delivery, reducing transfer failure rates by **35%**.
- Migrated legacy asynchronous code to **Swift Concurrency** (`async/await`, `actors`) from **GCD**, reducing concurrency-related crash rates by **40%** in long-running daemon processes.
- Optimized **Core Data** telemetry persistence via model redesign, indexing, and batch operations, reducing lookup latency by **50%**, and built analytics pipelines surfacing device health and network usage diagnostics to IT administrators.
- Built an internal diagnostics utility adopted by **20+ engineers and QA** for real-time log parsing, job payload inspection, and live agent state inspection, shortening incident triage.

#### iOS Developer Intern

February 2023 – June 2023

- Contributed to PKG/DMG creation, **code signing**, and **notarization**; implemented secure local storage using **Core Data**, **UserDefaults**, and **Keychain**; wrote macOS **shell scripts** and worked in **Agile** sprints with **Jira** and **SonarQube**.

## PROJECTS

### SureMDM Agent for macOS *Enterprise endpoint management platform — shipping product*

42Gears

- Production macOS device-management agent covering app lifecycle management, security policy enforcement, network monitoring, and remote administration. 42gears.com — SureMDM Agent for macOS

### CodeForge — DSA & Interview Prep IDE *Swift, SwiftUI, Swift Concurrency, JavaScriptCore*

Live on the App Store

- Designed, built, and published a fully offline native **macOS and iOS** coding-practice IDE as sole developer, with a sandboxed dual-path execution engine (subprocess and **JavaScriptCore**) and no server dependency. *[ADD IF TRUE: downloads / rating]*
- apps.apple.com/app/id6787703716 | github.com/sachin6174/SwiftForge

### Secure Text *JavaScript, Chrome Extension APIs*

Published on Chrome Web Store

- Chrome extension providing Base64 and Base32 text encoding and decoding via Chrome storage APIs. chromewebstore.google.com/detail/secure-text

## TECHNICAL SKILLS

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**Endpoint & Systems:** MDM/UEM, endpoint management, LaunchDaemon/launchd, XPC inter-process communication, privileged helper tools, process daemonization, macOS status-bar agents, device monitoring, remote administration, PKG/DMG packaging, code signing, notarization, entitlements, provisioning profiles

**Security & Privacy:** CryptoKit, Keychain Services, encryption at rest, Just-In-Time (JIT) privilege escalation, least privilege, policy enforcement, geofencing/time-fencing compliance, secure local storage, security hardening, SDK modernization

**Networking:** NetworkExtension (NEFilterProvider, NEAppProxyProvider), network traffic filtering, application proxying, per-app data usage tracking, REST APIs, URLSession, JSON/Codable

**Languages:** Swift, Objective-C, C++, Bash/Shell scripting, JavaScript, SQL

**iOS & macOS Frameworks:** SwiftUI, UIKit, Foundation, Combine, Swift Concurrency (async/await, actors, Sendable), GCD/DispatchQueue, Core Data, SwiftData, Core Location, Core ML

**Architecture & Quality:** VIPER, MVVM, Clean Architecture, SOLID principles, Protocol-Oriented Programming, Dependency Injection, XCTest, XCUITest, TDD, Playwright, Appium, SonarQube, Xcode Instruments, Code Review

**Build & Automation:** Xcode, SPM, CocoaPods, Fastlane, Git, GitLab CI/CD, TestFlight, App Store Connect, Jira, Agile/Scrum, Local LLMs, AI agents, release automation

## EDUCATION

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**Chandigarh Group of Colleges, Landran**

*Bachelor of Engineering in Computer Science and Engineering — CGPA: 7.9/10*

**August 2019 – May 2023**

*Mohali, Punjab, India*

## ACHIEVEMENTS & CERTIFICATIONS

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- **System Design** — Coding Ninjas, February 2022: scalability, load balancing, database design, and microservices architecture.
- **Data Structures in C++** — Coding Ninjas, April 2022.
- **Google Cloud Agentic AI Hackathon 2025** — top 700 teams out of 57,000+ developers.